

Results

1 Results

1.0.1 Distribution of Risk Ratios Across Decisions

Out of the 6400 sampled universes, 1487 statistical models did not converge, with 1102 log-binomial models, 221 generalised estimating equations (GEE) and 164 logistic regression analyses resulting in non-convergence. The other three models converged in all universes. Across the other decisions, NAs in the results were distributed equally. In total, 4913 universes successfully resulted in risk ratios.

Across all 4913 specifications, RRs ranged from 0.71 to 2.11, indicating that the effect of marriage on the risk of a cardiovascular event ranged from negative to positive within this sample of the full multiverse. All mean RRs per decision option can be found in Table 2 (Section 2). The largest RR variation was found in the statistical model decision, where the mean RR ranged from 0.83 (Cox PH) to 1.12 (GEE) and in the adjustment set, where the mean RR ranged from 0.83 (for $ADJ \in \{\text{Smoke} + \text{Sex}\}$ or $\{\text{Country} + \text{Sex}\}$ or $\{\text{Sex}\}$) to 1.07 (for $ADJ \in \{\text{Age} + \text{Smoke} + \text{Physical activity}\}$).

As shown in Figure 1, the effect estimates and their significance varied across the 4913 universes. Table ?@tbl-RRsensitivity summarises the sensitivity of results to each analytical decision, by showing the range of the mean RR across decision options and the percentage of significant results across options. Model choice and adjustment sets had the largest impact on effect estimates (RR range: 0.83–1.12 and 0.83–1.07, respectively), while missing data handling and exclusion criteria had minimal impact (RR range: 0.94–0.95 for both).

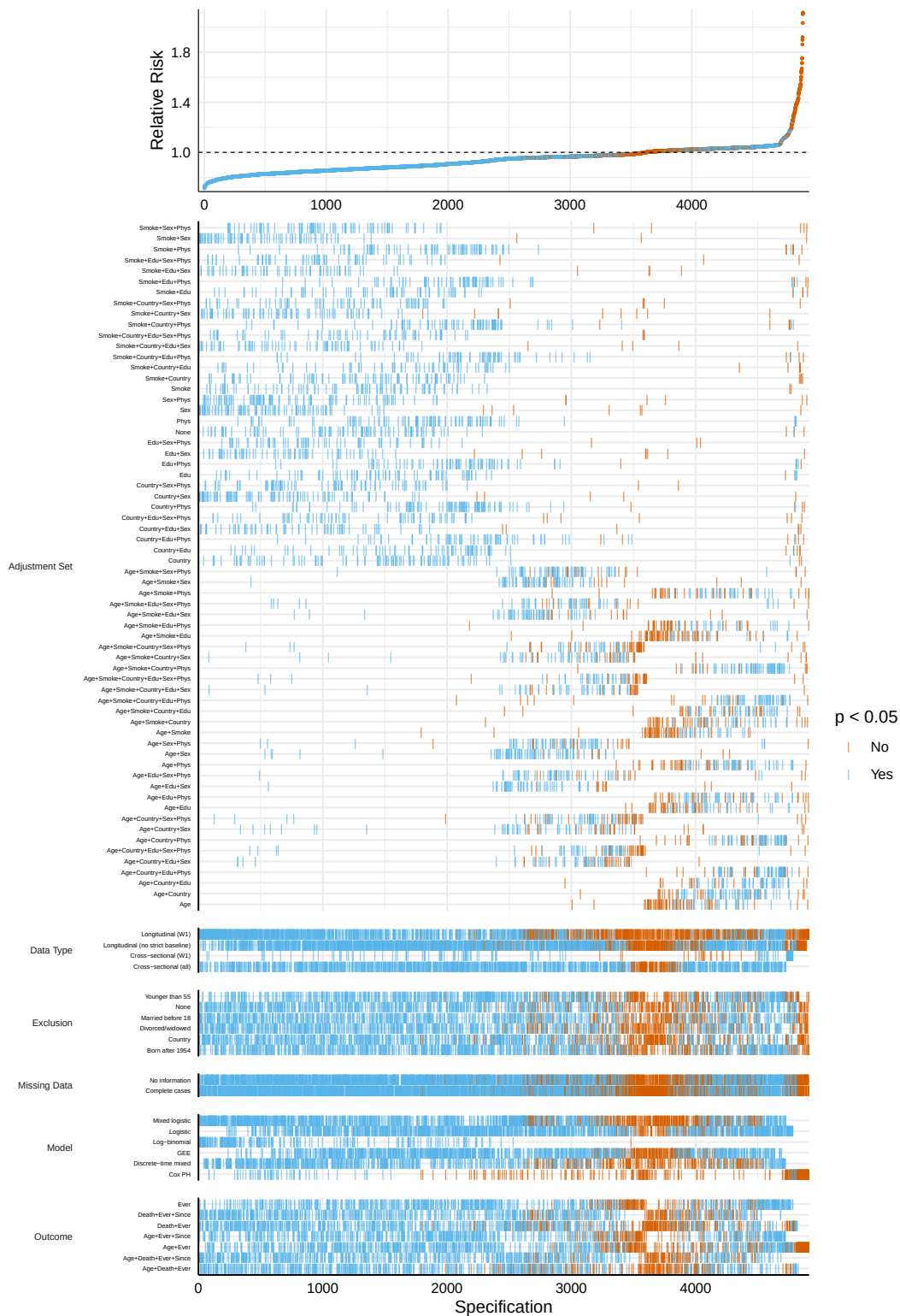


Figure 1: Specification curve multiverse results

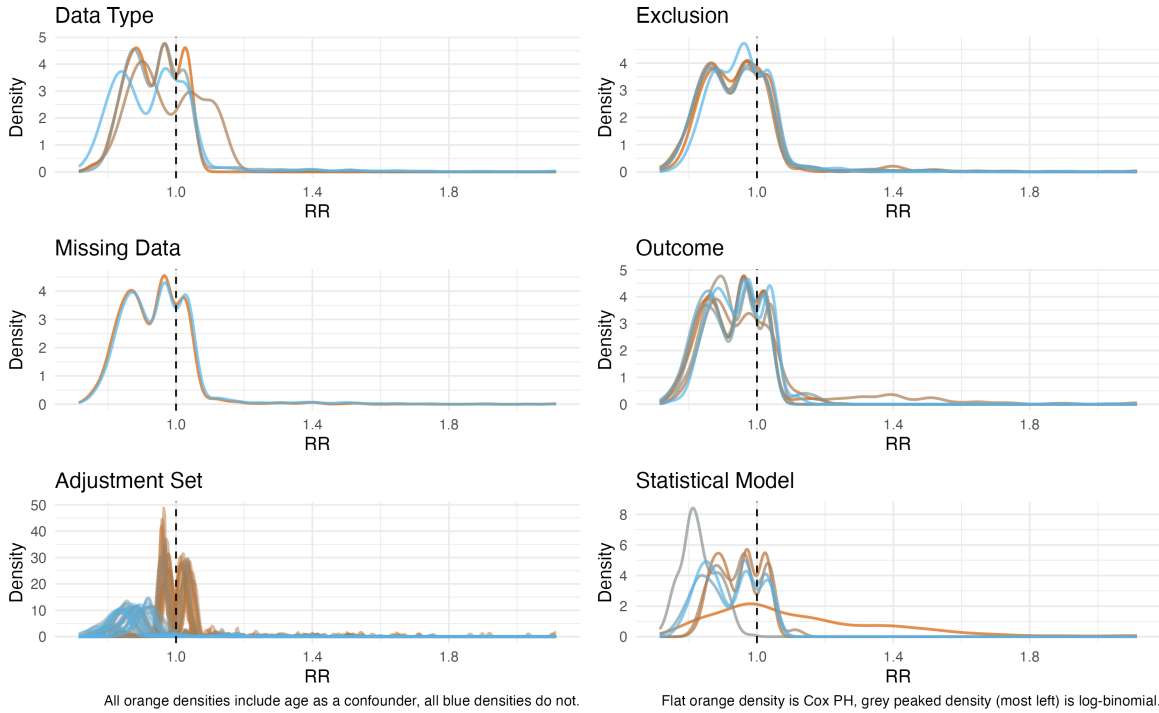
Table 1: K-sample Anderson-Darling tests for distributional differences across analytical decision options.

Decision	k	AD	T.AD	p-value
Data Type	4	73.10	53.17	< .001
Missing Data	2	5.48	5.88	0.002
Exclusion Criteria	6	28.16	13.62	< .001
Outcome	7	104.60	52.91	< .001
Statistical Model	6	511.54	297.73	< .001
Adjustment Set	64	2379.80	385.94	< .001

Note. k = number of options; T.AD = standardised Anderson-Darling statistic.

Density plots of RR per analytical decision

Each panel shows densities across decision options



Note: Dashed line indicates RR = 1

Figure 2: Densities of RR across decisions and options

2 Appendix

Table 2: RRs, their range and percentage of significant RRs across decisions and their options.

Decision	Option	Universes	Mean RR	RR Range	% Sig
Statistical Model	Log-binomial	232	0.83	[0.71, 0.98]	100%
	Mixed effects logistic regression	1278	0.92	[0.73, 1.07]	77%
	Discrete-time mixed effects	832	0.92	[0.76, 1.07]	74%
	GEE	1052	0.95	[0.8, 1.06]	86%
	Logistic regression	1188	0.95	[0.8, 1.14]	93%
	Cox PH	331	1.12	[0.77, 2.11]	24%
Data Type	Cross-sectional (all waves)	1237	0.93	[0.74, 1.07]	91%
	Longitudinal (wave 1)	1669	0.94	[0.71, 2.11]	68%
	Longitudinal (no strict baseline)	1869	0.95	[0.75, 1.54]	82%
	Cross-sectional (wave 1)	138	0.97	[0.77, 1.14]	86%
Missing Data	Complete cases	2584	0.94	[0.72, 2.11]	79%
	No information	2329	0.95	[0.71, 2.11]	80%
Exclusion Criteria	Country	833	0.94	[0.73, 1.86]	76%
	Divorced/widowed	819	0.94	[0.73, 1.75]	78%
	Married before 18	778	0.94	[0.71, 1.71]	81%
	None	804	0.94	[0.73, 1.9]	80%
	Born after 1954	886	0.95	[0.74, 2.11]	83%
	Younger than 55	793	0.95	[0.76, 2.1]	79%
Outcome	Age event + Death + Event ever + Since	644	0.92	[0.73, 1.06]	84%
	Death + Event ever + Since	656	0.92	[0.71, 1.06]	84%
	Age event + Death + Event ever	707	0.93	[0.75, 1.24]	76%
	Age event + Event ever + Since	617	0.93	[0.74, 1.07]	85%
	Death + Event ever	761	0.93	[0.72, 1.2]	76%
	Event ever	695	0.95	[0.73, 1.14]	83%
	Age event + Event ever	833	1.00	[0.77, 2.11]	72%

(continued)

Decision	Option	Universes	Mean RR	RR Range	% Sig
Adjustment Set	Smoke + Sex	81	0.83	[0.74, 1.55]	96%
	Country + Sex	100	0.83	[0.73, 1.18]	95%
	Sex	100	0.83	[0.71, 1.34]	93%
	Smoke + Edu + Sex	65	0.84	[0.75, 1.02]	94%
	Smoke + Country + Edu + Sex	74	0.85	[0.74, 1.23]	93%
	Smoke + Country + Sex	85	0.85	[0.75, 1.05]	89%
	Smoke + Edu + Sex + Phys	66	0.86	[0.77, 1.3]	95%
	Country + Edu + Sex	73	0.86	[0.75, 1.34]	92%
	Country + Sex + Phys	79	0.86	[0.77, 1.02]	97%
	Edu + Sex	75	0.86	[0.76, 1.47]	89%
	Edu + Sex + Phys	68	0.86	[0.8, 1.02]	96%
	Sex + Phys	100	0.86	[0.76, 1.6]	94%
	Smoke	86	0.87	[0.77, 1.12]	98%
	Smoke + Country + Edu + Sex + Phys	63	0.87	[0.78, 0.99]	95%
	Smoke + Country + Sex + Phys	72	0.87	[0.77, 1.32]	92%
	None	100	0.87	[0.76, 1.43]	97%
	Smoke + Sex + Phys	84	0.88	[0.8, 1.6]	94%
	Country	100	0.88	[0.76, 1.36]	98%
	Smoke + Country	87	0.89	[0.78, 1.39]	95%
	Country + Edu + Sex + Phys	73	0.89	[0.79, 1.38]	90%
	Edu	78	0.89	[0.79, 1.19]	99%
	Phys	100	0.90	[0.81, 1.18]	100%
	Smoke + Country + Edu	66	0.91	[0.79, 1.32]	92%
	Smoke + Edu	67	0.91	[0.79, 1.65]	94%
	Smoke + Phys	87	0.92	[0.81, 1.38]	93%
	Country + Edu	75	0.92	[0.81, 1.31]	93%
	Country + Phys	100	0.92	[0.82, 1.48]	96%
	Smoke + Country + Phys	84	0.93	[0.82, 1.14]	94%
	Edu + Phys	79	0.93	[0.83, 1.36]	97%
	Smoke + Country + Edu + Phys	69	0.94	[0.83, 1.41]	94%
	Smoke + Edu + Phys	75	0.94	[0.84, 1.75]	95%
	Age + Sex	82	0.95	[0.83, 1.03]	88%
	Country + Edu + Phys	72	0.95	[0.83, 1.42]	92%
	Age + Smoke + Sex	74	0.96	[0.82, 1.36]	82%
	Age + Country + Edu + Sex	69	0.96	[0.81, 1.02]	65%
	Age + Edu + Sex	60	0.96	[0.83, 1.13]	80%
	Age + Smoke + Country + Edu + Sex	62	0.97	[0.78, 1.46]	68%
	Age + Smoke + Edu + Sex	68	0.97	[0.84, 1.65]	78%
	Age + Country + Sex	77	0.97	[0.81, 1.43]	68%
	Age + Sex + Phys	79	0.97	[0.83, 1.9]	78%
	Age + Smoke + Country + Edu + Sex + Phys	61	0.98	[0.83, 1.4]	52%
	Age + Smoke + Edu + Sex + Phys	62	0.98	[0.83, 1.92]	79%
	Age + Smoke + Sex + Phys	79	0.98	[0.86, 1.5]	77%
	Age + Country + Edu + Sex + Phys	80	0.98	[0.82, 1.5]	56%
	Age + Country + Sex + Phys	86	0.98	[0.79, 1.51]	57%
	Age + Edu + Sex + Phys	65	0.98	[0.83, 1.51]	77%
	Age + Smoke + Country + Sex	75	0.99	[0.77, 1.54]	68%
	Age + Smoke + Country + Sex + Phys	72	0.99	[0.82, 1.49]	43%
	Age + Smoke	75	1.01	[0.89, 1.04]	33%
	Age + Smoke + Edu	80	1.02	[0.95, 1.2]	21%
	Age	87	1.03	[0.97, 1.67]	33%
	Age + Smoke + Country + Edu	67	1.04	[0.95, 1.3]	60%
	Age + Smoke + Edu + Phys	61	1.04	[0.92, 2.1]	43%
	Age + Country	87	1.04	[0.97, 1.44]	74%
	Age + Edu	71	1.04	[0.98, 2.03]	44%
	Age + Phys	85	1.04	[0.98, 1.52]	60%
	Age + Smoke + Country	73	1.05	[0.93, 1.66]	42%
	Age + Smoke + Country + Edu + Phys	66	1.05	[0.91, 1.71]	76%
	Age + Country + Edu	63	1.05	[0.96, 1.58]	75%
	Age + Country + Phys	77	1.05	[0.99, 1.71]	70%
	Age + Smoke + Country + Phys	78	1.06	[0.97, 1.64]	83%
	Age + Country + Edu + Phys	71	1.06	[1, 1.75]	77%
	Age + Edu + Phys	70	1.06	[1.01, 1.86]	59%
	Age + Smoke + Phys	68	1.07	[0.96, 2.11]	56%